

Solving Quadratic Equations by Extracting Square Roots

Jonathan R. Bacolod

A **quadratic equation** can sometimes be solved
by isolating the squared expression first.

Square Root Property

If an equation can be written in the form

$$x^2 = a$$

then apply the **square root property**:

$$x = \pm\sqrt{a}$$

Extending the Property

If the equation has the form

$$(x - h)^2 = a$$

then

$$x - h = \pm\sqrt{a}$$

and solve for x .

Case: Two Real Solutions

If $a > 0$, the equation has **two real solutions**.

$$x^2 = a, \quad a > 0 \implies x = \pm\sqrt{a}$$

Case: One Real Solution

If $a = 0$, the equation has **one real solution**.

$$x^2 = 0 \implies x = 0$$

Case: No Real Solutions

If $a < 0$, the equation has **no real solutions**.

$$x^2 = a, \quad a < 0 \implies \text{no real } x$$

Check Your Work

Always **check your solutions** by substituting them into the original equation.

Summary: Value of a and Solutions

Value of a	Number of Solutions
$a > 0$	Two real solutions
$a = 0$	One real solution
$a < 0$	No real solutions

Examples

Example 1: Two Real Solutions

Solve

$$x^2 = 25$$

$$x = \pm\sqrt{25}$$

$$x = 5 \quad \text{or} \quad x = -5$$

Example 2: One Real Solution

Solve

$$(x - 2)^2 = 0$$

$$x - 2 = \pm\sqrt{0}$$

$$x = 2$$

Example 3: No Real Solutions

Solve

$$(x + 1)^2 = -4$$

Since $a = -4 < 0$, there is no real square root.

This equation has **no real solutions**.

Practice Exercises

Practice 1

Solve.

$$x^2 = 49$$

Practice 2

Solve.

$$x^2 = 144$$

Practice 3

Solve.

$$x^2 = \frac{81}{16}$$

Practice 4

Solve.

$$(x - 4)^2 = 25$$

Practice 5

Solve.

$$(x + 3)^2 = 64$$

Practice 6

Solve.

$$3x^2 = 108$$

Practice 7

Solve.

$$2(x - 1)^2 = 98$$

Practice 8

Solve.

$$(2x + 1)^2 = 49$$

Practice 9

Solve.

$$(3x - 6)^2 = 144$$

Activities

Activity 1

Solve.

$$x^2 = 64$$

Activity 2

Solve.

$$x^2 = 225$$

Activity 3

Solve.

$$x^2 = \frac{121}{25}$$

Activity 4

Solve.

$$(x - 6)^2 = 16$$

Activity 5

Solve.

$$(x + 5)^2 = 81$$

Activity 6

Solve.

$$5x^2 = 320$$

Activity 7

Solve.

$$4(x + 2)^2 = 144$$

Activity 8

Solve.

$$(2x - 3)^2 = 81$$

Activity 9

Solve.

$$(4x + 8)^2 = 256$$

Thank you!